An Exponential Mortality Bound for Finite Real Matrix Monoids

github.com/egilburg/aimath

8 September 2026
Manuscript revision 2026-09-08-r2

Abstract

Let $n > 0$, and let a family of real $n \times n$ matrices generate a finite entire product monoid containing zero. We prove that zero is represented by a word of length at most

$$b(n) = 2^{n-1} + (2^{n-1} - 1)\frac{n(n+1)}{2} = \Theta(n^2 2^n).$$

For rational matrices this improves the bound 3^{n^2} of Kiefer and Ryzhikov (2026), Theorem 7.1, under the same finite-product promise and without irreducibility. It also improves Almeida and Steinberg's (2009) universal rational mortality bound $(2n-1)^{n^2}-1$ for $n > 1$, Theorem 5.6, supplying the exponential alternative in their Question 5.7; the polynomial alternative remains unsettled. The proof combines established finite-group averaging, compressed returns and symmetric-matrix lifting into a short-word test for rank descent. An accompanying Lean development proves the real theorem and rational specialization. Discoveries presented here were made through the use of artificial intelligence.

1 The theorem and its comparison

For an indexing set A and matrices $M_a \in M_n(\mathbb{R})$, write

$$M_w = M_{a_1} \cdots M_{a_m} \quad (w = a_1 \cdots a_m), \qquad M_\epsilon = I_n.$$

Let $S = \{M_w : w \in A^*\}$. Mortality means $0 \in S$. Our hypothesis is that S itself is finite, not merely that the family of generators is finite.

Theorem 1.1. *Let $n \geq 1$. If the entire monoid $S \subseteq M_n(\mathbb{R})$ generated by a family $(M_a)_{a \in A}$ is finite and contains zero, then some word $z \in A^*$ satisfies*

$$M_z = 0, \qquad |z| \leq b(n) := 2^{n-1} + (2^{n-1} - 1)\frac{n(n+1)}{2}.$$

No finiteness assumption on A is needed.

The same bound holds over $\mathbb{Q}$, with the same dimension and word. Finiteness of the whole product monoid is essential to this proof. We do not address unrestricted matrix mortality or the length needed to represent every element of S.

1.1 Prior bounds and established methods

Almeida–Steinberg [1, Definitions 2.1 and 5.1] introduced the universal mortality-function framework used below. Their Theorem 5.6 gives 1 at $n = 1$ and

$$(2n-1)^{n^2} - 1 \qquad (n > 1).$$

Kiefer–Ryzhikov [2, Theorem 7.1] give 3^{n^2} for mortality in finite rational matrix semigroups. Although their title emphasizes irreducibility, that mortality theorem has no irreducibility hypothesis. Adjoining an identity preserves finiteness and aligns a finite product semigroup with our monoid convention.

The rational consequence therefore changes the exponent from quadratic in n to $n + O(\log n)$. The formulas avoid ambiguity about exponential terminology. Kiefer–Ryzhikov's full cardinality and arbitrary minimum-rank conclusions remain separate; our theorem is zero-specific.

Finite-group averaging, compression to return semigroups, and orbit-span stabilization are established techniques. Protasov–Voynov [4, Sections 7 and 9.1] provide particularly relevant invariant-form and symmetric-matrix lifting antecedents, including the dimension $n(n+1)/2$. The contribution considered here is their combination into a scalar test which vanishes on invertible compressed returns, is nonzero at a global zero product, and yields an explicit mortality-length recurrence.

For nonnegative matrices, the established support-based bound $2^n - 1$ does not require finiteness of the arithmetic product monoid [5]. It is stronger on that subclass. The signed real and rational setting is thus material to the present comparison.

2 Short-word detection

Lemma 2.1. *Let V have dimension $d \geq 1$, let $x \in V$, let $T_a \in \operatorname{End}(V)$, and let $\ell \in V^*$. If $\ell(T_w x) \neq 0$ for some word, then this holds for a word of length at most $d - 1$.*

Proof. Put $L_m = \operatorname{span}\{T_w x : |w| \leq m\}$. Then

$$L_{m+1} = L_m + \sum_{a \in A} T_a L_m.$$

A plateau makes L_m invariant under every T_a, so it is permanent. The hypothesis implies $x \neq 0$, hence $\dim L_0 = 1$. The full reachable span is therefore already L_{d-1}. If ℓ vanished on all orbit vectors of length at most $d - 1$, it would vanish on the whole reachable span, a contradiction. ◻

This classical span argument works for an arbitrary indexing set of operators.

Write $N = n(n+1)/2$. On $\mathbb{R} \oplus \operatorname{Sym}_n(\mathbb{R})$, of dimension $1 + N$, define

$$\mathcal{T}_a(c, B) = (c, M_a B M_a^\top).$$

These maps are linear and, with the stated word convention,

$$\mathcal{T}_w(c, B) = (c, M_w B M_w^\top).$$

Applying Lemma 2.1 to a linear observation gives:

Corollary 2.2. *For a symmetric B, scalar c, and linear functional f on $M_n(\mathbb{R})$, if*

$$c - f(M_w B M_w^\top) \neq 0$$

for some word, it is nonzero for a word of length at most N.

The scalar coordinate expresses the affine defect linearly. The word bound is N, one less than the lifted dimension.

3 An invariant form for compressed returns

Lemma 3.1. *Let $\mathcal{R} \subseteq M_r(\mathbb{R})$ be finite and multiplicatively closed, with $r > 0$. There is a symmetric matrix Q with $\operatorname{tr} Q > 0$ such that*

$$KQK^{\mathsf{T}} = Q \quad \textit{for every invertible } K \in \mathcal{R}.$$

Proof. If there are no invertible members, use $Q = I_r$. Otherwise the invertible members form a finite group G. Indeed, repetition among positive powers of an invertible member shows that its identity and inverse belong to $\mathcal{R}$; multiplicative closure is inherited.

Set $Q = \sum_{g \in G} gg^{\mathsf{T}}$. It is symmetric; every summand has nonnegative trace and the identity contributes $r > 0$. Left multiplication by $K \in G$ permutes the group, so

$$KQK^{\mathsf{T}} = \sum_{g \in G} (Kg)(Kg)^{\mathsf{T}} = Q.$$

$\square$

Although the group average is positive definite, the proof below needs only positive trace and invariance. No inverse of Q is used.

Choose a nonzero $H \in S$ with rank r, and a rank factorization

$$H = UV, \quad U \in M_{n,r}(\mathbb{R}), \quad V \in M_{r,n}(\mathbb{R}),$$

where U is injective and V is surjective. For $Y \in S$, put $K_Y = VYU$. The compressed set $\mathcal{R}_H = \{K_Y : Y \in S\}$ is finite and multiplicatively closed:

$$K_Y K_Z = V(YHZ)U, \qquad YHZ \in S.$$

It need not contain I_r before invertible members are considered. Also,

$$HYH = UK_YV, \qquad \operatorname{rank}(HYH) = \operatorname{rank} K_Y. \tag{1}$$

The rank equality uses the injection and surjection in the factorization.

4 A quadratic-length middle word

Proposition 4.1. *Under the hypotheses of Theorem 1.1, for every word h with $H = M_h \neq 0$, there is a word w with*

$$|w| \leq N, \qquad \operatorname{rank}(M_{hwh}) < \operatorname{rank} H.$$

Proof. Apply Lemma 3.1 to $\mathcal{R}_H$, and define

$$\delta(Y) = \operatorname{tr} Q - \operatorname{tr}(K_Y Q K_Y^{\mathsf{T}}).$$

Invertible K_Y have $\delta(Y) = 0$. Hence a nonzero defect implies that K_Y is singular and, by (1), that $\operatorname{rank}(HYH) < r$. No converse is used: some singular returns may also have zero defect.

Mortality supplies nontriviality, since $\delta(0) = \operatorname{tr} Q > 0$. Put

$$B = UQU^{\mathsf{T}} \in \operatorname{Sym}_n(\mathbb{R}), \qquad f(Z) = \operatorname{tr}(VZV^{\mathsf{T}}).$$

Multiplication gives

$$f(YBY^{\mathsf{T}}) = \operatorname{tr}(VYUQU^{\mathsf{T}}Y^{\mathsf{T}}V^{\mathsf{T}}) = \operatorname{tr}(K_Y Q K_Y^{\mathsf{T}}).$$

Corollary 2.2, with $c = \operatorname{tr} Q$, produces a word w of length at most N for which $\delta(M_w) \neq 0$. This forces the required rank decrease. $\square$

The quadratic length uses the original dimension n, not the compressed rank. Both outer copies of h remain in the conclusion and must be counted.

5 Global descent and universal mortality

Proof of Theorem 1.1. As $n > 0$, the empty word is not zero. A zero product contains a singular generator, since products of invertible matrices are invertible. Start with that one-letter word h_0, of rank at most $n-1$.

While its matrix is nonzero, use Proposition 4.1 and replace h_i by $h_{i+1} = h_i w_i h_i$. Ranks strictly decrease, while lengths obey

$$L_{i+1} \le 2L_i + N, \qquad L_0 = 1.$$

Induction gives $L_i \le 2^i + (2^i - 1)N$. After at most $n-1$ rank decreases the matrix is zero, with length at most $b(n)$. An initially zero letter also satisfies the bound. In particular, $b(1) = 1$. $\square$

Corollary 5.1. *The same bound holds for rational matrices, without changing the dimension or the witnessing word.*

Proof. The entrywise embedding $\mathbb{Q} \hookrightarrow \mathbb{R}$ preserves products, preserves finiteness of the product range, and reflects equality to zero. Apply Theorem 1.1 and retain the word. $\square$

Corollary 5.2. *The function b is a universal rational mortality function in the sense of Almeida–Steinberg: for every finite monoid F, every monoid representation $\rho : F \to M_n(\mathbb{Q})$ whose image contains zero, and every generating set A of F, some word of length at most $b(n)$ has zero image.*

Proof. The matrices $\rho(a)$ generate the finite image $\rho(F)$. Apply Corollary 5.1. $\square$

Since $b(n) = 2^{n+O(\log n)}$, this supplies the exponential alternative in [1, Question 5.7]. It is the stated mathematical implication, not a claim of historical first resolution. The polynomial alternative, a sharp mortality threshold and quadratic rational Rystsov remain unresolved here.

6 Formal verification and provenance

The accompanying `lean/` directory contains a self-contained development for this paper, `Publication.lean`, `Audit.lean`, and dependency pins. The environment is Lean 4.33.1 and Mathlib commit `0df444a360eaa60ab8c11dca51a86af692955474`, credited in [3, 6]. The included `verification/` supplement supplies build logs, exact statement and axiom audits, source hashes, and verification provenance; its `README.txt` explains reproduction and `SHA256SUMS` checks the proof files. The manuscript-to-formal correspondence is recorded in `CLAIM_MAP.txt`. This revision extracts the needed lemmas into subject-specific modules; the supplement records their correspondence to the earlier proof and the fresh build checks.

In namespace `FiniteMonoidMortality`, the principal declarations are:

- `exists_short_zero_word_of_finite_real_monoid`;
- `exists_short_zero_word_of_finite_rational_monoid`;
- `exists_short_rank_decreasing_sandwich_of_finite`.

The formal hypotheses are $0 < n$, finiteness of the entire word-product range, and existence of a zero word. There is no hidden finite-alphabet premise. Endpoint axioms are limited to `propext`, `Classical.choice`, and `Quot.sound`. Corollary 5.2 applies the rational endpoint to a representation image; this is an exposition-level corollary rather than a separately named Lean theorem.

Documented research process. The AI investigation explored whether local rank compression could yield a quadratic global bound, then retained the cost of both outer word copies. Finite return groups and the symmetric-matrix lift supplied a nonzero scalar observation, allowing a short middle word. Starting with a singular generator sharpened the resulting exponential bound. The formal version simplified an earlier inverse-form expression to a trace defect and fixed a direct word-product convention. Later literature checks separated the general signed-matrix result from stronger nonnegative subclasses.

The theorem is existential: it gives no cost bound for computing return groups, no polynomial-time algorithm for arbitrary real input, and no bound for representing every monoid element. Nonnegative mortality lower bounds do not automatically apply under the finite-monoid promise. Optimality and global priority remain unestablished. Reports should identify the version and statement through `CORRECTIONS.txt`.

AI origin and review disclosure. All original mathematical contributions and discoveries presented here were produced by AI; established mathematics is credited separately. The publisher-supplied default attribution is OpenAI GPT-6 Astra. Exact model provenance for individual discovery, formalization and drafting stages was not retained or could not be recovered, so this is not runtime-verified model telemetry. The publisher selected and organized the research and publication, does not claim subject-matter expertise, and has not professionally reviewed the mathematics. Claims are supplied as-is, without a publisher assurance of correctness, completeness or originality. Readers should perform whatever independent mathematical and computational verification is necessary before relying on them. Lean verification concerns encoded statements and their assumptions, not the accuracy of every sentence or historical priority.

References

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